Prevention and Screening Strategies for Colon Cancer

Prevention and early detection of colon cancer are central to reducing incidence and mortality. The American Gastroenterological Association, the US Preventive Services Task Force, and the American College of Radiology recommend that average-risk adults initiate colorectal cancer screening at age 45, reflecting the rising incidence of early-onset disease. Screening should continue through age 75, with individualized decisions for those aged 76 to 85 based on comorbidities and prior screening history.

[1-3]

For average-risk individuals, recommended modalities include colonoscopy every 10 years, annual fecal immunochemical testing (FIT), multitarget stool DNA-FIT every 1 to 3 years, flexible sigmoidoscopy every 5 to 10 years (with or without FIT), and CT colonography every 5 years.

[1-4]

Blood-based tests are not recommended as first-line options due to lower sensitivity and higher cost.

[3]

For individuals with a first-degree relative diagnosed with colorectal cancer or advanced adenoma before age 60, or with two first-degree relatives at any age, colonoscopy should begin at age 40 or 10 years before the earliest diagnosis in the family, whichever comes first, and be repeated every 5 years.

[2][5]

For those with a family history diagnosed at age 60 or older, colonoscopy may be performed every 5 to 10 years.

[2][5]

High-risk groups, including those with hereditary syndromes such as Lynch syndrome (hereditary nonpolyposis colorectal cancer, HNPCC) or familial adenomatous polyposis (FAP), require more intensive surveillance. For Lynch syndrome, colonoscopy is recommended every 1 to 2 years starting at age 20 to 25, or 2 to 5 years earlier than the youngest diagnosis in the family.

[6]

For FAP, flexible sigmoidoscopy or colonoscopy should begin at age 10 to 12 and be repeated every 1 to 2 years until colectomy.

[6]

Pediatric populations do not undergo routine screening unless they have a hereditary syndrome. Children with FAP should begin endoscopic surveillance at age 10 to 12, and those with Lynch syndrome at age 20 to 25, or earlier based on family history.

[6]

For patients with inflammatory bowel disease involving the colon, surveillance colonoscopy with biopsies should begin 8 to 10 years after onset of colitis and be repeated every 1 to 3 years.

[6]

Primary prevention focuses on modifiable risk factors. Maintaining a healthy weight, engaging in regular moderate to vigorous physical activity, consuming a diet rich in fruits, vegetables, and whole grains, and limiting red and processed meats, tobacco, and alcohol are all associated with reduced risk.

[3][7]

Aspirin chemoprevention may be considered in select high-risk adults aged 50 to 59 with a 10-year cardiovascular risk of 10% or greater and no increased bleeding risk.

[3]

Table 2 from Jayakrishnan and Ng published in JAMA provides a comprehensive summary of screening recommendations and modifiable risk factors for early-onset gastrointestinal cancers, including colorectal cancer, and quantifies the impact of various risk factors and interventions.

[8]

Table 2. Screening Recommendations and Modifiable Risk Factors for Early-Onset Gastrointestinal Cancers[a]

	Type of early-onset gastrointestinal cancer		
	Colorectal	**Pancreatic**	**Esophagogastric**
Screening recommendations	While some countries, such as Austria and Italy, start screening at age 40 years,[18] and the US at age 45 years (USPSTF guidelines),[5] most other countries start screening at age 50 years. Recommendations vary by region and risk factors.[b]	Both US guidelines (American College of Gastroenterology,[19] American Gastroenterology Assocation[20]) and international guidelines (International Cancer of the Pancreas Screening Consortium[21]) recommend screening only for high-risk individuals (eg, those with inherited genetic syndromes, familial pancreatic cancer), with age varying by risk factors (eg, begin at age 50 years or 10 years younger than the initial age of familial onset; age 40 years in *CKDN2A* and *PRSS1* gene variant carriers with hereditary pancreatitis; age 35 years with Peutz-Jeghers syndrome)[20]	Routine EGD screening for early-onset esophagogastric cancer is not currently performed in most countries. However, in South Korea, biennial screening for gastric cancer with EGD or upper gastrointestinal series begins at age 40 years.[22] The American College of Gastroenterology recommends a single screening EGD for individuals with chronic gastroesophageal reflux disease symptoms and at least 3 of the following risk factors: male sex, age >50 years, White race, tobacco use, obesity, and family history of Barrett esophagus or esophageal adenocarcinoma in a first-degree relative.[23] Taipei Global Consensus guidelines endorse mass screening and eradication of *Helicobacter pylori* in individuals aged 20-40 years in regions with a high incidence or high risk of gastric cancer.[24]
Associated risk factors[c]			
Diet			
Sugar-sweetened beverages	RR, 2.18 (95% CI, 1.10-4.35; $P = .02$) for sugar-sweetened beverage consumption ≥2 servings per day vs <1 serving per week[25]	Not available	Beef and canned, smoked, and salted food: OR, 2.1 (95% CI, 1.3-2.8)[26]
Western dietary pattern	OR, 1.67 (95% CI, 1.18-2.37; $P < .03$) for high-risk adenomas (highest vs lowest quintile for Western diet score)[27]	Not available	Not available
Obesity (body mass index >30)[d]	OR, 1.88 (95% CI, 1.40-2.54)[28]	OR, 1.85 (95% CI, 1.19-2.88)[29]	OR, 1.7 (95% CI, 1.5-2.0)[26]
Alcohol consumption	RR, 1.71 (95% CI, 1.62-1.80) for highest defined category in studies compared with never drinkers[30]	Not available	OR, 2.0 (95% CI, 1.3-2.6) for alcohol consumption vs none[26]
Physical activity	Sedentary behavior: RR, 1.69 (95% CI, 1.07-2.67; $P = .03$); television viewing >14 h/wk vs <7 h/wk[31]	Physical activity (protective): HR, 0.25 (95% CI, 0.07-0.93) for active vs inactive participants per study criteria[32]	Not available
Smoking	OR, 1.33 (95% CI, 1.17-1.52) for current smoking vs nonsmoking[33]	HR, 1.84 (95% CI, 1.29-2.62) for ever smoking vs nonsmoking[29]	OR, 1.5 (95% CI, 0.6-2.3) for smoking vs nonsmoking[26]
MASLD (fatty liver index >30)[e]	HR, 1.14 (95% CI, 1.06-1.22)[34]	HR, 1.23 (95% CI, 1.09-1.40)[34]	HR, 1.14 (95% CI, 1.06-1.24)[34]
Infection	Not available	Not available	*H pylori*: OR, 2.3 (95% CI, 1.4-3.2)[26]
Others			
Hyperlipidemia	RR, 1.61 (95% CI, 1.22-2.13)[30]	Not available	Not available
Breastfed as infant	OR, 1.46 (95% CI, 1.16-1.83) for high-risk adenomas among those breastfed in infancy vs not breastfed[35]	Not available	Not available
Vitamin D intake	HR, 0.49 (95% CI, 0.26-0.83) for total vitamin D intake including dietary and supplemental intake (protective) of ≥450 IU/d vs <300 IU/d[36]	Not available	Not available

Abbreviations: EGD, esophagogastroduodenoscopy; HR, hazard ratio; MASLD, metabolic dysfunction–associated steatotic liver disease; OR, odds ratio; RR, relative risk; USPSTF, US Preventive Services Task Force.

[a] Preference was given to high-quality meta-analyses and large prospective cohort studies where available. Definitions of terms are included where available. Early onset is defined as an age at diagnosis <50 years and later onset is defined as an age at diagnosis ≥50 years unless otherwise noted.

[b] Individuals with a personal history of colorectal cancer, inflammatory bowel disease, hereditary colorectal cancer syndromes, other colorectal cancer–predisposing conditions, or a family history of colorectal cancer.[5]

[c] The ORs, RRs, and HRs in this section apply to groups with vs without exposure to risk factors unless otherwise specified. Definitions per the cited studies are included where available. The risk factors are ordered by descending numerical value for early-onset colorectal cancer and prioritizing those with the most data across cancers.

[d] Body mass index is calculated as weight in kilograms divided by height in meters squared.

[e] Referred to as "nonalcoholic fatty liver disease" in the study.[34]

Table 2. Screening Recommendations and Modifiable Risk Factors for Early-Onset Gastrointestinal Cancersa

Early-Onset Gastrointestinal Cancers. JAMA. July 16, 2025.

Content used under license from the JAMA Network®

© American Medical Association

Screening adherence remains suboptimal, with only about 68.8% of eligible US adults undergoing recommended screening.

[9]

Barriers include lack of knowledge, limited access, and reluctance to undergo screening, with lower rates among underinsured, low-income, and minority populations.

[1][9]

Organized outreach, mailed FIT programs, patient navigation, and culturally tailored education have been shown to significantly increase screening rates and reduce disparities.

[10-15]

Diagnosis, Staging, and Molecular Profiling

The diagnostic workup for colon cancer begins with a thorough history and physical examination, including assessment of symptoms (rectal bleeding, change in bowel habits, abdominal pain, weight loss), family history, and risk factors.

[2][16]

Laboratory evaluation includes baseline carcinoembryonic antigen (CEA) measurement, which is useful for monitoring but not diagnostic in isolation.

[16]

Colonoscopy with biopsy is the gold standard for diagnosis, allowing for direct visualization, biopsy, and polypectomy.

[1][16-17]

In cases where colonoscopy is incomplete or contraindicated, CT colonography or flexible sigmoidoscopy may be considered, but abnormal findings require follow-up colonoscopy.

[1][17]

Staging is performed according to the American Joint Committee on Cancer (AJCC) Tumor, Node, Metastasis (TNM) system, which incorporates tumor depth (T), nodal involvement (N), and distant metastasis (M).

[16]

The 8th edition of the AJCC system further refines metastatic categories and highlights the importance of lymphovascular and perineural invasion, microsatellite instability (MSI), tumor budding, and KRAS, NRAS, and BRAF mutations in treatment considerations.

[16]

Cross-sectional imaging with contrast-enhanced CT of the chest, abdomen, and pelvis is recommended for staging and to assess for metastatic disease, particularly in the liver and lungs.

[16]

PET/CT is reserved for cases with indeterminate findings or suspected metastatic disease not well characterized by conventional imaging.

[16]

Molecular profiling is now standard for all newly diagnosed colon cancers. Universal testing for MSI or mismatch repair (MMR) protein expression is recommended to identify Lynch syndrome and for prognostic and therapeutic stratification.

[8-9][16]

Germline multigene panel testing is advised for all patients diagnosed under age 50, as pathogenic variants are present in approximately one in six early-onset cases, with half involving MMR genes.

[8-9][18]

Additional molecular markers, including KRAS, NRAS, and BRAF mutations, are assessed in metastatic disease to guide targeted therapy selection.

[19-24]

Next-generation sequencing (NGS) is increasingly advocated for early and routine use to identify rare actionable alterations and to facilitate response monitoring through circulating tumor DNA (ctDNA) analysis.

[22-25]

Pediatric colon cancer requires prompt colonoscopic evaluation and genetic counseling for symptomatic children or those with a strong family history. Immunohistochemical staining and molecular testing for MMR deficiency and MSI are recommended, as these findings have implications for prognosis and treatment.

[26-28]

Pediatric tumors show a higher frequency of RNF43 mutations and CDK6 amplification, with less frequent APC mutations than adults, suggesting age-related differences in oncogenic pathways.

[27-28]

Treatment Approaches Across All Stages

Stage-Specific and Molecularly Guided Therapy

Treatment of colon cancer is stage-specific and increasingly personalized. For stage I disease, surgical resection alone is generally curative, and adjuvant therapy is not recommended.

[16][19][29]

For stage II disease, adjuvant chemotherapy is reserved for patients with high-risk features, such as T4 tumors, poor differentiation, lymphovascular or perineural invasion, obstruction, perforation, or inadequate lymph node sampling.

[16][19][29]

Fluoropyrimidine-based chemotherapy (capecitabine or 5-FU/leucovorin) is recommended for high-risk stage II patients, but oxaliplatin is generally not added unless additional risk factors are present.

[19][30]

For stage III disease, adjuvant chemotherapy is standard following surgical resection. The National Comprehensive Cancer Network (NCCN) and the American Society of Clinical Oncology (ASCO) recommend CAPEOX (capecitabine plus oxaliplatin) for 3 months or FOLFOX (5-FU/leucovorin plus oxaliplatin) for 3 to 6 months in low-risk stage III disease,

and CAPEOX for 3 to 6 months or FOLFOX for 6 months in high-risk stage III disease, based on the IDEA trial.

[19][30]

The addition of oxaliplatin confers a survival benefit, particularly in stage IIIA patients, except for certain subgroups (e.g., T1N2A) where benefit is less clear.

[30]

Table 2 from Nors et al published in JAMA Oncology provides quantitative data on recurrence risk by stage, which is critical for tailoring surveillance and adjuvant therapy.

[31]

Table 2. Risk of Recurrence in Patients With Stages I to III Colorectal Cancer

Characteristic	Cumulative incidence function, % (95% CI)[a]			sHR (95% CI)[b]
	1 y	3 y	5 y	
Colon cancer				
UICC stage I				
2004-2008	3.4 (2.4-4.6)	11.5 (9.6-13.5)	16.3 (14.0-18.6)	1 [Reference]
2009-2013	4.1 (3.1-5.4)	9.5 (7.9-11.3)	12.5 (10.7-14.5)	0.75 (0.60-0.94)
2014-2019	1.5 (1.1-2.0)	4.4 (3.7-5.2)	6.8 (5.9-7.8)	0.37 (0.30-0.45)
UICC stage II				
2004-2008	7.2 (6.3-8.2)	17.0 (15.6-18.4)	21.9 (20.4-23.4)	1 [Reference]
2009-2013	5.4 (4.6-6.2)	13.6 (12.4-14.8)	17.0 (15.7-18.3)	0.76 (0.68-0.85)
2014-2019	3.6 (3.0-4.2)	9.7 (8.8-10.6)	11.6 (10.6-12.6)	0.50 (0.44-0.56)
UICC stage III				
2004-2008	12.2 (10.8-13.7)	29.8 (27.8-31.8)	35.3 (33.2-37.4)	1 [Reference]
2009-2013	11.6 (10.2-13.0)	26.9 (25.0-28.8)	31.1 (29.1-33.1)	0.85 (0.77-0.95)
2014-2019	9.8 (8.9-10.8)	21.8 (20.4-23.1)	24.6 (23.2-26.1)	0.65 (0.59-0.71)
Rectal cancer				
UICC stage I				
2004-2008	5.4 (4.0-7.1)	13.9 (11.6-16.4)	19.9 (17.2-22.8)	1 [Reference]
2009-2013	5.1 (3.8-6.7)	13.1 (11.0-15.3)	17.1 (14.8-19.6)	0.84 (0.68-1.05)
2014-2019	2.7 (2.0-3.5)	7.3 (6.2-8.5)	9.5 (8.2-10.9)	0.44 (0.36-0.55)
UICC stage II				
2004-2008	8.2 (6.8-9.9)	20.8 (18.5-23.1)	25.8 (23.4-28.2)	1 [Reference]
2009-2013	7.6 (6.1-9.2)	18.9 (16.7-21.2)	22.7 (20.4-25.2)	0.87 (0.74-1.02)
2014-2019	5.1 (4.0-6.4)	14.8 (13.0-16.8)	18.4 (16.3-20.6)	0.66 (0.56-0.79)
UICC stage III				
2004-2008	12.4 (10.6-14.2)	32.3 (29.7-34.8)	38.7 (36.0-41.3)	1 [Reference]
2009-2013	11.9 (10.2-13.8)	28.1 (25.6-30.6)	32.6 (30.0-35.1)	0.83 (0.73-0.94)
2014-2019	11.1 (9.6-12.7)	24.0 (21.9-26.1)	28.8 (26.6-31.1)	0.70 (0.62-0.80)

Abbreviations: sHR, subdistribution hazard ratio; UICC, Union for International Cancer Control.

[a] Cumulative incidence function of colorectal cancer recurrence treating death and second primary cancer (other than colorectal cancer or nonmelanoma skin cancer) as competing events.

[b] sHRs calculated by Fine-Gray regression adjusted for age, sex, and Charlson Comorbidity Index score.

Table 2. Risk of Recurrence in Patients With Stages I to III Colorectal Cancer

Incidence of Recurrence and Time to Recurrence in Stage I to III Colorectal Cancer: A Nationwide Danish Cohort Study. JAMA Oncol. December 31, 2023.

Content used under license from the JAMA Network®

© American Medical Association

For locally advanced colon cancer (T4 or bulky tumors), neoadjuvant chemotherapy (FOLFOX or CAPOX) is increasingly utilized to downstage tumors, facilitate margin-negative resection, and reduce recurrence rates, as demonstrated in the FOxTROT, OPTICAL, and NeoCol trials.

[32-33]

For patients with deficient mismatch repair (dMMR) tumors, neoadjuvant immunotherapy (e.g., immune checkpoint inhibitors) has shown promising pathologic response rates, as exemplified by the NICHE trial.

[32-33]

Management of metastatic colon cancer (stage IV) is highly individualized and guided by molecular profiling, disease burden, and resectability. The ASCO guideline recommends doublet or triplet chemotherapy (FOLFOX, FOLFIRI, CAPOX) as first-line therapy for unresectable metastatic disease, often in combination with targeted agents.

[20-21]

Tumor sidedness and molecular features (RAS, BRAF, MSI/MMR status) are critical in selecting biologic therapies.

[19-21][23-24]

For RAS wild-type, left-sided tumors, chemotherapy plus anti-EGFR therapy (cetuximab or panitumumab) is recommended; for RAS wild-type, right-sided tumors, chemotherapy plus anti-VEGF therapy (bevacizumab) is preferred.

[20-21][23]

For BRAF V600E mutant tumors, the combination of encorafenib and cetuximab is FDA-approved and recommended after prior therapy.

[20-21][23][34]

For dMMR/MSI-H tumors, immunotherapy with pembrolizumab or nivolumab is recommended as first-line or subsequent therapy, with improved progression-free and overall survival demonstrated in the KEYNOTE-177 and CheckMate 142 trials.

[20-21][35]

Figure 1 from Biller and Schrag published in JAMA illustrates prototypical treatment trajectories for patients with unresectable metastatic colorectal cancer according to molecular subtype, including median survival times and recommended regimens.

[21]

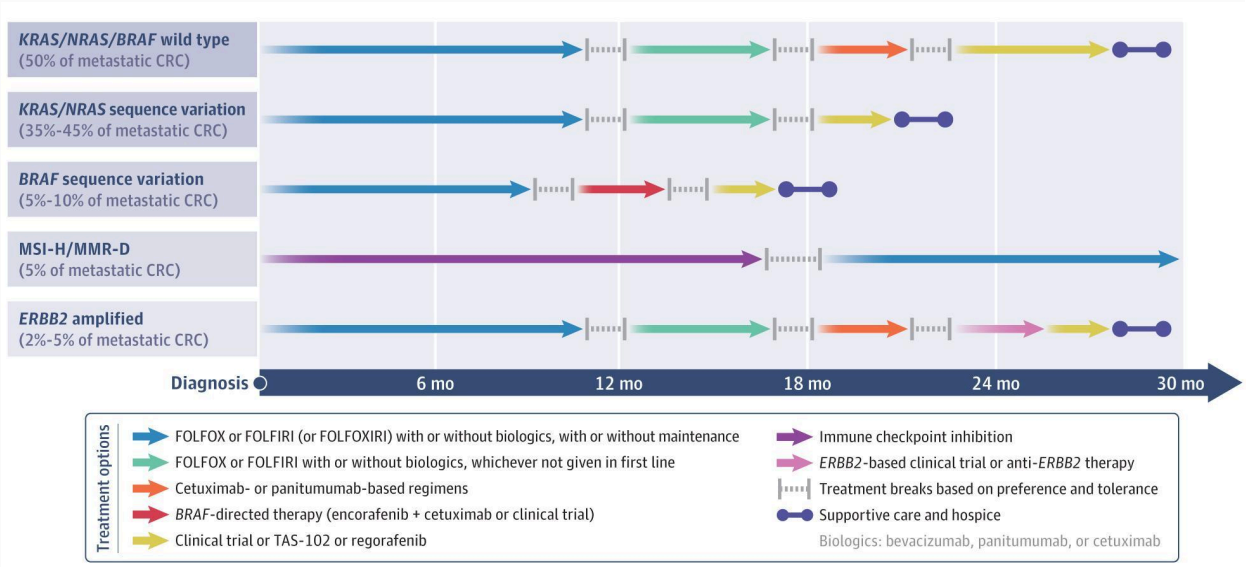

Figure 1. Prototypical Treatment Trajectories for Patients With Unresectable Metastatic Colorectal Cancer According to Molecular Subtype

Diagnosis and Treatment of Metastatic Colorectal Cancer: A Review. JAMA. February 15, 2021.

Content used under license from the JAMA Network®

© American Medical Association

For patients with potentially resectable metastatic disease, perioperative chemotherapy (total of six months, split between neoadjuvant and adjuvant) is recommended, with surgery for both primary and metastatic sites performed either synchronously or in a staged approach.

[36]

Multidisciplinary team management is essential for optimal sequencing and decision-making.

[20][36]

Radiation therapy is not routinely used in colon cancer except in select cases of locally advanced disease where it may facilitate resection or palliate symptoms.

[33][36]

For rectal cancer, neoadjuvant chemoradiotherapy is standard for locally advanced tumors, but in colon cancer, its role is limited and should be considered on a case-by-case basis.

[33][36]

Emerging therapies include novel targeted agents (KRAS G12C inhibitors, anti-HER2 therapies, WEE1 and PARP inhibitors), immunotherapy combinations, and cell-based approaches (CAR-T, CAR-NK, dendritic cell vaccines, oncolytic viruses).

[22-24][37-40]

Next-generation sequencing and ctDNA analysis are increasingly used to guide therapy selection and monitor response.

[22-25]

Pediatric and Hereditary Syndromes

Pediatric colon cancer is rare and typically presents at an advanced stage with aggressive histology (mucinous adenocarcinoma, signet ring cell carcinoma).

[27][41-44]

Management generally follows adult protocols, with surgical resection and combination chemotherapy (5-FU, oxaliplatin, irinotecan) as the mainstays.

[27][43][45-46]

Targeted agents such as bevacizumab and cetuximab may be added in higher-risk or metastatic cases, although access to these drugs and clinical trials is limited for pediatric patients.

[44-46]

Pediatric tumors show distinct molecular profiles, including higher frequency of RNF43 mutations and CDK6 amplification, and less frequent APC mutations than adults.

[27-28]

Germline mutations in MMR genes (Lynch syndrome, CMMRD) are identified in a substantial proportion of pediatric cases, and universal genetic counseling and multigene panel testing are recommended for all patients under 50 years of age.

[8-9][18][26]

Hereditary syndromes such as Lynch syndrome and FAP require tailored management, including consideration of extended surgical resection and intensified surveillance.

[8-9][16][18][47-48]

Universal germline genetic testing is recommended for early-onset and high-risk patients to guide therapy and familial risk assessment.

[8-9][18][47-48]

Surveillance, Survivorship, and Management of Late Effects

Post-treatment surveillance is essential for early detection of recurrence and management of late effects. The American Society of Colon and Rectal Surgeons and the American Cancer Society recommend structured surveillance protocols tailored to stage and risk.

[49-50]

For stage II and III adults, office visits and CEA testing are recommended every 3 to 6 months for the first two years, then every 6 to 12 months for the next three years, for a total of five years.

[49-51]

Annual CT scans of the chest, abdomen, and pelvis are recommended for up to five years in high-risk patients, with less frequent imaging for lower-risk patients.

[49-51]

Colonoscopy is recommended one year after resection, then at three and five-year intervals if no advanced neoplasia is detected.

[49-52]

For rectal cancer survivors, proctoscopy or flexible sigmoidoscopy every 6 months for five years is recommended to detect local recurrence.

[52]

For stage I patients, less intensive surveillance is appropriate, with colonoscopy at one year post-resection and subsequent intervals as above; routine CEA and CT imaging are generally not recommended.

[31][50][52]

Pediatric patients should follow adult surveillance protocols, with intensified endoscopic surveillance for hereditary syndromes.

[49-50][52]

For Lynch syndrome, colonoscopy every 1 to 2 years starting at age 20 to 25 is recommended for life; for FAP, flexible sigmoidoscopy or colonoscopy every 1 to 2 years starting at age 10 to 12 is recommended until colectomy, with subsequent surveillance of the remaining rectum or ileal pouch.

[6][52]

Survivorship care plans (SCPs) are recommended for all patients, including a treatment summary, follow-up schedule, and information about late and long-term effects.

[49-50][53]

Table 1 from Shapiro published in The New England Journal of Medicine provides site-specific surveillance recommendations for cancer survivors, including colorectal cancer, and outlines the frequency and modalities for monitoring.

[54]

Table 1. Suggested Site-Specific Surveillance Recommendations for Cancer Survivors.*

Disease Site	Recommendations	Comments
Head and neck cancer[5]†	Physical examination every 1–3 mo for 1 yr, then every 2–6 mo for 2–5 yr and annually after 5 yr Baseline imaging 6 mo after completion of treatment Indirect laryngoscopy performed by an ENT physician periodically Low-dose CT scans for lung-cancer screening, indicated for persons at high risk because of a history of smoking	If new or persistent symptoms develop, imaging is performed as appropriate to the clinical situation
Breast cancer[6]†	Physical examination every 3–4 mo for 3 yr, then every 6 mo for 2 yr, and annually after 5 yr‡ Breast imaging annually	Imaging or measurement of tumor markers is not indicated in women without symptoms; if new or persistent symptoms develop, imaging is indicated as appropriate to the clinical situation
Prostate cancer[7]§	Digital rectal examination annually for 5 yr PSA test every 6–12 mo for 5 yr	Imaging in men without symptoms is not indicated; if new or persistent symptoms develop, imaging is indicated as appropriate to the clinical situation
Colorectal cancer[10]§	Physical examination and CEA test every 3–6 mo for 5 yr CT imaging of chest, abdomen, and pelvis annually for 3 yr Colonoscopy annually for 6 yr after surgery	If new or persistent symptoms develop, imaging is indicated as appropriate to the clinical situation
Non–small-cell lung cancer[12]	History taking and physical examination every 3–6 mo for 1–2 yr, then annually for 3–5+ yr Low-dose axial CT scanning every 6 mo for 1–2 yr, then annually for 3–5+ yr¶	If new or persistent symptoms develop, imaging is indicated as appropriate to the clinical situation
Testicular cancer[13]	Follow-up guidelines, which depend on histologic features (e.g., seminoma or nonseminoma) and stage	If new or persistent symptoms develop, imaging is indicated as appropriate to the clinical situation
Gynecologic cancer[14]	Follow-up guidelines, which depend on histologic features (e.g., endometrial, cervical, or ovarian cancer) and stage	If new or persistent symptoms develop, imaging is indicated as appropriate to the clinical situation
Lymphoma[15]	Follow-up guidelines, which depend on histologic features (diffuse large lymphoma, follicular lymphoma, or Hodgkin's disease) and stage	If new or persistent symptoms develop, imaging is indicated as appropriate to the clinical situation

* Regarding cancer treated with bone marrow transplantation,[16] virtually every organ system may be affected by high-dose chemotherapy with allogeneic or autologous bone marrow transplantation. Specific surveillance guidelines for long-term and late effects of childhood cancers depend on organ site and exposure risk; in children who receive high-dose chemotherapy with allogeneic bone marrow transplantation, almost every organ system may be affected[11,15] (https://childrensoncologygroup.org/index.php/survivorshipguidelines). CEA denotes carcinoembryonic antigen, CT computed tomography, DXA dual-energy x-ray absorptiometry, ENT ear, nose, and throat, and PSA prostate-specific antigen.

† The American Society of Clinical Oncology practice guidelines are available at www.asco.org/practice-guidelines/cancer-care-initiatives/prevention-survivorship/survivorship-compendium.

‡ The recommendations are for women receiving antiestrogen therapy.

§ American Cancer Society surveillance guidelines for survivors of prostate and colorectal cancers are available at www.cancer.org/health-care-professionals/american-cancer-society-survivorship-guidelines/prostate-cancer-survivorship-care-guideline.html and www.cancer.org/health-care-professionals/american-cancer-society-survivorship-guidelines/colorectal-cancer-survivorship-care-guidelines.html, respectively.

¶ Surveillance with low-dose CT for more than 5 years is controversial.

Table 1. Suggested Site-Specific Surveillance Recommendations for Cancer Survivors.*

Cancer Survivorship. N Engl J Med. December 19, 2018.

Used under license from The New England Journal of Medicine.

Management of late and long-term effects is multidisciplinary. Peripheral neuropathy, particularly from oxaliplatin, affects up to 40% of survivors and may be managed with duloxetine, physical and occupational therapy, and exercise.

[49][54-56]

Bowel dysfunction, ostomy-related issues, urogenital and sexual dysfunction, fatigue, cognitive impairment, and chronic pain are common and require systematic assessment and intervention.

[49-50][54-60]

Immunotherapy-related adverse events (irAEs) may affect the gastrointestinal tract, endocrine system, skin, and liver, and require prompt recognition and management, often with corticosteroids or hormone replacement.

[54][58]

Psychosocial support, including screening for depression, anxiety, and fear of recurrence, is essential, with referral to mental health professionals as needed.

[49-50][58][60-61]

Exercise, cognitive rehabilitation, and lifestyle modification (smoking cessation, healthy diet, weight management) are recommended to improve quality of life and reduce recurrence risk.

[50][54][58-59][61]

Patient Counseling, Prognosis, and Addressing Disparities

Effective patient counseling requires clear, compassionate, and individualized communication, as recommended by the American Society of Clinical Oncology.

[62]

Patients and families frequently ask about causes and risk factors, symptoms, diagnosis and staging, treatment options, prognosis, family risk and screening, lifestyle modification, side effects, genetic testing, and post-treatment surveillance.

[1-3][5][8-9][16][63]

Evidence-based answers should be grounded in the latest guidelines and research, with quantitative data provided wherever possible.

Prognosis is determined by a constellation of factors, with stage at diagnosis being the most important. Five-year survival rates are highest for localized disease (>90%), decrease for regional disease (72–85%), and are lowest for distant metastatic disease (~8–20%).

[64-67]

Age, tumor grade, lymph node ratio, tumor location, comorbidities, and molecular features further modulate survival.

[68-75]

Right-sided colon cancers generally have worse outcomes than left-sided cancers, and pediatric patients have poorer prognosis than adults, with five-year survival around 40% compared to 60% in adults.

[27][41][43-44]

Table 2 from Nors et al published in JAMA Oncology, previously displayed, provides quantitative recurrence risks by stage.

[31]

Health disparities in colon cancer incidence, screening, and outcomes are well documented, particularly among Black, American Indian, Alaska Native, and low-income populations.

[1][9][76-77]

The primary drivers are inequities in access to and utilization of screening and treatment, rather than genetic differences.

[1][9][76]

Evidence-based strategies to address disparities include organized mailed FIT outreach, patient navigation, culturally tailored education, reduction of structural barriers, and multicomponent interventions integrated into primary care and community settings.

[10-15][78]

These approaches have been shown to increase screening rates, reduce disparities, and improve outcomes in underserved populations.

Emerging research is rapidly advancing the field. Molecular profiling, next-generation sequencing, and ctDNA analysis are increasingly used to guide therapy and monitor response.

[22-25]

Novel targeted agents, immunotherapy combinations, and cell-based therapies are under investigation, with promising results in select molecular subgroups.

[22-24][35][37-40]

Pediatric and early-onset cases are being included in clinical trials, and there is a critical need for age-specific strategies and collaborative research to improve outcomes in these populations.

[27-28]

In summary, the management of colon cancer is highly stratified by risk, stage, molecular profile, and patient characteristics. Adherence to evidence-based guidelines for prevention, screening, diagnosis, treatment, surveillance, and survivorship is essential for optimizing outcomes. Quantitative protocols, multidisciplinary care, and patient-centered communication are central to high-quality care. Ongoing research and targeted interventions to address disparities will further improve outcomes for all patients with colon cancer.

References:

1.

Screening for Colorectal Cancer: US Preventive Services Task Force Recommendation Statement.

Davidson KW, Barry MJ, Mangione CM, et al.

JAMA. 2021;325(19):1965-1977. doi:10.1001/jama.2021.6238.

2.

AGA Clinical Practice Update on Risk Stratification for Colorectal Cancer Screening and Post-Polypectomy Surveillance: Expert Review.

Issaka RB, Chan AT, Gupta S.

Gastroenterology. 2023;165(5):1280-1291. doi:10.1053/j.gastro.2023.06.033.

Practice Guideline

3.

Colorectal Cancer Screening and Prevention.

Sur DKC, Brown PC.

American Family Physician. 2025;112(3):278-283.

4.

How Would You Screen This Patient for Colorectal Cancer? : Grand Rounds Discussion From Beth Israel Deaconess Medical Center.

Burns RB, Mangione CM, Weinberg DS, Kanjee Z.

Annals of Internal Medicine. 2022;175(10):1452-1461. doi:10.7326/M22-1961.

5.

Screening for Colorectal Cancer and Evolving Issues for Physicians and Patients: A Review.

Lieberman D, Ladabaum U, Cruz-Correa M, et al.

JAMA. 2016;316(20):2135-2145. doi:10.1001/jama.2016.17418.

6.

ACR Appropriateness Criteria® Colorectal Cancer Screening: 2024 Update.

Thomas DE, Horvat N, Fowler KJ, et al.

Journal of the American College of Radiology : JACR. 2025;22(5S):S190-S201. doi:10.1016/j.jacr.2025.02.032.

Practice Guideline

7.

Colorectal Cancer: Epidemiology, Risk Factors, and Prevention.

Roshandel G, Ghasemi-Kebria F, Malekzadeh R.

Cancers. 2024;16(8):1530. doi:10.3390/cancers16081530.

8.

Early-Onset Gastrointestinal Cancers.

Jayakrishnan T, Ng K.

JAMA. 2025;:2836671. doi:10.1001/jama.2025.10218.

9.

Increasing Incidence of Early-Onset Colorectal Cancer.

Sinicrope FA.

The New England Journal of Medicine. 2022;386(16):1547-1558. doi:10.1056/NEJMra2200869.

10.

Colorectal Cancer Screening After Sequential Outreach Components in a Demographically Diverse Cohort.

Podmore C, Selby K, Jensen CD, et al.

JAMA Network Open. 2024;7(4):e245295. doi:10.1001/jamanetworkopen.2024.5295.

11.

Interventions to Increase Colorectal Cancer Screening Adherence in Low-Income Settings Within the United States: A Systematic Review and Meta-Analysis.

Rubin L, Okitondo C, Haines L, Ebell M.

Preventive Medicine. 2023;172:107522. doi:10.1016/j.ypmed.2023.107522.

12.

Centralized Colorectal Cancer Screening Outreach in Federally Qualified Health Centers: A Randomized Clinical Trial.

Reuland DS, O'Leary MC, Crockett SD, et al.

JAMA Network Open. 2024;7(11):e2446693. doi:10.1001/jamanetworkopen.2024.46693.

13.

ACG Clinical Guidelines: Colorectal Cancer Screening 2021.

Shaukat A, Kahi CJ, Burke CA, et al.

The American Journal of Gastroenterology. 2021;116(3):458-479. doi:10.14309/ajg.0000000000001122.

Practice Guideline

14.

Systematic Review of Randomised Controlled Trials on Interventions Aimed at Promoting Colorectal Cancer Screening Amongst Ethnic Minorities.

Abdul Latip SNB, Chen SE, Im YR, Zielinska AP, Pawa N.

Ethnicity & Health. 2023;28(5):661-695. doi:10.1080/13557858.2022.2139815.

15.

Mi-Care: Comparing Three Evidence-Based Interventions to Promote Colorectal Cancer Screening Among Ethnic Minorities Within Three Different Clinical Contexts.

Watson KS, Tossas KY, San Miguel Y, et al.

International Journal of Environmental Research and Public Health. 2023;20(22):7049. doi:10.3390/ijerph20227049.

16.

The American Society of Colon and Rectal Surgeons Clinical Practice Guidelines for the Management of Colon Cancer.

Vogel JD, Felder SI, Bhama AR, et al.

Diseases of the Colon and Rectum. 2022;65(2):148-177. doi:10.1097/DCR.0000000000002323.

Practice Guideline

17.

Multitarget Stool RNA Test for Colorectal Cancer Screening.

Barnell EK, Wurtzler EM, La Rocca J, et al.

JAMA. 2023;330(18):1760-1768. doi:10.1001/jama.2023.22231.

18.

Delphi Initiative for Early-Onset Colorectal Cancer (DIRECt) International Management Guidelines.

Cavestro GM, Mannucci A, Balaguer F, et al.

Clinical Gastroenterology and Hepatology : The Official Clinical Practice Journal of the American Gastroenterological Association. 2023;21(3):581-603.e33. doi:10.1016/j.cgh.2022.12.006.

Practice Guideline

19.

Optimizing Adjuvant Therapy for Localized Colon Cancer and Treatment Selection in Advanced Colorectal Cancer.

Grothey A, Venook AP.

Journal of the National Comprehensive Cancer Network : JNCCN. 2018;16(5S):611-615. doi:10.6004/jnccn.2018.0038.

20.

Treatment of Metastatic Colorectal Cancer: ASCO Guideline.

Morris VK, Kennedy EB, Baxter NN, et al.

Journal of Clinical Oncology : Official Journal of the American Society of Clinical Oncology. 2023;41(3):678-700. doi:10.1200/JCO.22.01690.

Practice Guideline

21.

Diagnosis and Treatment of Metastatic Colorectal Cancer: A Review.

Biller LH, Schrag D.

JAMA. 2021;325(7):669-685. doi:10.1001/jama.2021.0106.

22.

An Update on Pharmacotherapies for Colorectal Cancer: 2023 and Beyond.

Zhu M, Benson AB.

Expert Opinion on Pharmacotherapy. 2024;:1-9. doi:10.1080/14656566.2024.2304654.

23.

Current Advances in Targeted Therapy for Metastatic Colorectal Cancer - Clinical Translation and Future Directions.

Johnson D, Chee CE, Wong W, et al.

Cancer Treatment Reviews. 2024;125:102700. doi:10.1016/j.ctrv.2024.102700.

24.

Precision Medicine for Metastatic Colorectal Cancer: Where Do We Stand?.

Underwood PW, Pawlik TM.

Cancers. 2024;16(22):3870. doi:10.3390/cancers16223870.

25.

Colorectal Cancer: A Comprehensive Review of Carcinogenesis, Diagnosis, and Novel Strategies for Classified Treatments.

Abedizadeh R, Majidi F, Khorasani HR, Abedi H, Sabour D.

Cancer Metastasis Reviews. 2024;43(2):729-753. doi:10.1007/s10555-023-10158-3.

26.

Clinical and Genetic Characteristics of Pediatric Colorectal Cancer.

Shen Q, Zhou Y, Liu X, et al.

Pediatric Blood & Cancer. 2025;72(4):e31569. doi:10.1002/pbc.31569.

27.

Recent Advances in Pediatric Colorectal Cancer: A Systematic Review.

Dhoat N, Sharma S.

Pediatric Surgery International. 2025;41(1):174. doi:10.1007/s00383-025-06076-w.

28.

Molecular Profiling of Pediatric and Young Adult Colorectal Cancer Reveals a Distinct Genomic Landscapes and Potential Therapeutic Avenues.

Busico A, Gasparini P, Rausa E, et al.

Scientific Reports. 2024;14(1):13138. doi:10.1038/s41598-024-64149-7.

29.

Treatment of Patients With Early-Stage Colorectal Cancer: ASCO Resource-Stratified Guideline.

Costas-Chavarri A, Nandakumar G, Temin S, et al.

Journal of Global Oncology. 2019;5:1-19. doi:10.1200/JGO.18.00214.

Practice Guideline

30.

Trends in the treatment of stage IIB/C and IIIA colorectal cancer: A National Cancer Database (NCDB) analysis.

Budhathoki P, Karki U, Pokharel N, et al.

Journal of Clinical Oncology. 2025;43(Suppl 16):e15658. doi:10.1200/JCO.2025.43.16_suppl.e15658.

31.

Incidence of Recurrence and Time to Recurrence in Stage I to III Colorectal Cancer: A Nationwide Danish Cohort Study.

Nors J, Iversen LH, Erichsen R, Gotschalck KA, Andersen CL.

JAMA Oncology. 2024;10(1):54-62. doi:10.1001/jamaoncol.2023.5098.

32.

Role of Neoadjuvant Therapies in Locally Advanced Colon Cancer.

de Castria TB, Lenz G, Valagni G, Kim RD.

Chinese Medical Journal. 2025;:00029330-990000000-01667. doi:10.1097/CM9.0000000000003756.

33.

Neoadjuvant Immunotherapy for Colorectal Cancer: Right Regimens, Right Patients, Right Directions?.

Zhu J, Lian J, Xu B, et al.

Frontiers in Immunology. 2023;14:1120684. doi:10.3389/fimmu.2023.1120684.

34.

FDA Orange Book.

FDA Orange Book

35.

Nivolumab plus Ipilimumab in Microsatellite-Instability–High Metastatic Colorectal Cancer.

Andre T, Elez E, Van Cutsem E, et al.

The New England Journal of Medicine. 2024;391(21):2014-2026. doi:10.1056/NEJMoa2402141.

36.

Treatment of Patients With Late-Stage Colorectal Cancer: ASCO Resource-Stratified Guideline.

Chiorean EG, Nandakumar G, Fadelu T, et al.

JCO Global Oncology. 2020;6:414-438. doi:10.1200/JGO.19.00367.

Practice Guideline

37.

Current and Emerging Treatment Paradigms in Colorectal Cancer: Integrating Hallmarks of Cancer.

Salva de Torres C, Baraibar I, Saoudi González N, et al.

International Journal of Molecular Sciences. 2024;25(13):6967. doi:10.3390/ijms25136967.

38.

Current and Future Trends of Colorectal Cancer Treatment: Exploring Advances in Immunotherapy.

Nikolouzakis TK, Chrysos E, Docea AO, et al.

Cancers. 2024;16(11):1995. doi:10.3390/cancers16111995.

39.

Emerging Frontiers in Colorectal Cancer Therapy: From Targeted Molecules to Immunomodulatory Breakthroughs and Cell-Based Approaches.

Fatemi N, Mirbahari SN, Tierling S, et al.

Digestive Diseases and Sciences. 2025;70(3):919-942. doi:10.1007/s10620-024-08774-2.

40.

Emerging New Treatments for Colon Cancer.

Adedayo SI, Riethmacher E.

Current Medicinal Chemistry. 2025;:CMC-EPUB-148431. doi:10.2174/0109298673372776250505155945.

41.

Distinct Features of Colorectal Cancer in Children and Adolescents: A Population-Based Study of 159 Cases.

Sultan I, Rodriguez-Galindo C, El-Taani H, et al.

Cancer. 2010;116(3):758-65. doi:10.1002/cncr.24777.

42.

Pediatric Gastric and Colorectal Cancers: Insights From a 16-Year Single-Center Experience.

Halalsheh H, Barbar M, AlAdwan M, Sultan I.

Pediatric Blood & Cancer. 2025;:e31860. doi:10.1002/pbc.31860.

43.

Colorectal Carcinoma in Pediatric Patients: A Comparison With Adult Tumors, Treatment and Outcomes From the National Cancer Database.

Poles GC, Clark DE, Mayo SW, et al.

Journal of Pediatric Surgery. 2016;51(7):1061-6. doi:10.1016/j.jpedsurg.2015.11.005.

44.

Sporadic Mucinous Adenocarcinoma of the Colon in an Adolescent Male: A Case Report and Review of the Literature.

Gaston F, Polite B, Ullenius S, Fitzgerald S, Bajaj M.

Journal of Pediatric Hematology/Oncology. 2025;47(5):219-223. doi:10.1097/MPH.0000000000003031.

45.

Management and Pharmacotherapy of Pediatric Colorectal Carcinoma: A Review.

Bergamaschi L, Chiaravalli S, Signoroni S, Di Bartolomeo M, Ferrari A.

Expert Opinion on Pharmacotherapy. 2023 Sep-Dec;24(14):1527-1535. doi:10.1080/14656566.2023.2230123.

46.

Epidemiology and Management Options for Colorectal Cancer in Children.

Saab R, Furman WL.

Paediatric Drugs. 2008;10(3):177-92. doi:10.2165/00148581-200810030-00006.

47.

NCCN Guidelines Insights: Genetic/Familial High-Risk Assessment: Colorectal, Version 2.2019.

Gupta S, Provenzale D, Llor X, et al.

Journal of the National Comprehensive Cancer Network : JNCCN. 2019;17(9):1032-1041. doi:10.6004/jnccn.2019.0044.

Practice Guideline

48.

ACG Clinical Guideline: Genetic Testing and Management of Hereditary Gastrointestinal Cancer Syndromes.

Syngal S, Brand RE, Church JM, et al.

The American Journal of Gastroenterology. 2015;110(2):223-62; quiz 263. doi:10.1038/ajg.2014.435.

Practice Guideline

49.

The American Society of Colon and Rectal Surgeons Clinical Practice Guidelines for the Surveillance and Survivorship Care of Patients After Curative Treatment of Colon and Rectal Cancer.

Hardiman KM, Felder SI, Friedman G, et al.

Diseases of the Colon and Rectum. 2021;64(5):517-533. doi:10.1097/DCR.0000000000001984.

Practice Guideline

50.

American Cancer Society Colorectal Cancer Survivorship Care Guidelines.

El-Shami K, Oeffinger KC, Erb NL, et al.

CA: A Cancer Journal for Clinicians. 2015 Nov-Dec;65(6):428-55. doi:10.3322/caac.21286.

Practice Guideline

51.

Care of the Colorectal Cancer Survivor.

Burgers K, Moore C, Bednash L.

American Family Physician. 2018;97(5):331-336.

52.

Colonoscopy Surveillance After Colorectal Cancer Resection: Recommendations of the US Multi-Society Task Force on Colorectal Cancer.

Kahi CJ, Boland CR, Dominitz JA, et al.

Gastrointestinal Endoscopy. 2016;83(3):489-98.e10. doi:10.1016/j.gie.2016.01.020.

Practice Guideline

53.

Cancer Survivorship Care in the United States at Facilities Accredited by the Commission on Cancer.

Stal J, Miller KA, Mullett TW, et al.

JAMA Network Open. 2024;7(7):e2418736. doi:10.1001/jamanetworkopen.2024.18736.

54.

Cancer Survivorship.

Shapiro CL.

The New England Journal of Medicine. 2018;379(25):2438-2450. doi:10.1056/NEJMra1712502.

55.

Management of Treatment-Related Sequelae Following Colorectal Cancer.

Haas S, Mikkelsen AH, Kronborg CJS, et al.

Colorectal Disease : The Official Journal of the Association of Coloproctology of Great Britain and Ireland. 2023;25(3):458-488. doi:10.1111/codi.16299.

56.

Symptom Experiences in Colorectal Cancer Survivors After Cancer Treatments: A Systematic Review and Meta-Analysis.

Han CJ, Yang GS, Syrjala K.

Cancer Nursing. 2020 May/Jun;43(3):E132-E158. doi:10.1097/NCC.0000000000000785.

57.

Non-Gastrointestinal Symptom Burden Following Colorectal Cancer Treatment-a Systematic Review.

Fernandes D, Nelson D, Ortega M, et al.

Supportive Care in Cancer : Official Journal of the Multinational Association of Supportive Care in Cancer. 2024;32(10):699. doi:10.1007/s00520-024-08903-7.

58.

Management of Common Clinical Problems Experienced by Survivors of Cancer.

Emery J, Butow P, Lai-Kwon J, et al.

Lancet (London, England). 2022;399(10334):1537-1550. doi:10.1016/S0140-6736(22)00242-2.

59.

Primary Care of Adult Cancer Survivors.

Carek S, Emerson JF, Patel J.

American Family Physician. 2024;110(1):37-44.

60.

How Consequences of Colorectal Cancer Treatment Are Managed: A Qualitative Study of Stakeholder Experiences About Supportive Care and Current Practices.

Rutherford C, Ju A, Kim B, et al.

Supportive Care in Cancer : Official Journal of the Multinational Association of Supportive Care in Cancer. 2023;31(5):255. doi:10.1007/s00520-023-07713-7.

61.

A Literature Review of Post-Treatment Survivorship Interventions for Colorectal Cancer Survivors and/or Their Caregivers.

Luo X, Li J, Chen M, et al.

Psycho-Oncology. 2021;30(6):807-817. doi:10.1002/pon.5657.

62.

Patient-Clinician Communication: American Society of Clinical Oncology Consensus Guideline.

Gilligan T, Coyle N, Frankel RM, et al.

Journal of Clinical Oncology : Official Journal of the American Society of Clinical Oncology. 2017;35(31):3618-3632. doi:10.1200/JCO.2017.75.2311.

Practice Guideline

63.

Navigating the Colorectal Cancer Maze: Unveiling Pathways to Diagnosis, Management, Pathophysiology and Prevention.

Al Kamzari KAM, Constantinou C.

Current Oncology Reports. 2025;:10.1007/s11912-025-01707-w. doi:10.1007/s11912-025-01707-w.

64.

Colon Cancer Survival Rates With the New American Joint Committee on Cancer Sixth Edition Staging.

O'Connell JB, Maggard MA, Ko CY.

Journal of the National Cancer Institute. 2004;96(19):1420-5. doi:10.1093/jnci/djh275.

65.

Colon and Rectal Cancer Survival in Seven High-Income Countries 2010-2014: Variation by Age and Stage at Diagnosis (The ICBP SURVMARK-2 Project).

Araghi M, Arnold M, Rutherford MJ, et al.

Gut. 2021;70(1):114-126. doi:10.1136/gutjnl-2020-320625.

66.

Age Disparities in Stage-Specific Colon Cancer Survival Across Seven Countries: An International Cancer Benchmarking Partnership SURVMARK-2 Population-Based Study.

Pilleron S, Charvat H, Araghi M, et al.

International Journal of Cancer. 2021;148(7):1575-1585. doi:10.1002/ijc.33326.

67.

Net Survival in Colon and Rectal Cancer by Stage According to Neoadjuvant Treatment. A French Population-Based Study.

Jooste V, Grosclaude P, Defossez G, et al.

Digestive and Liver Disease : Official Journal of the Italian Society of Gastroenterology and the Italian Association for the Study of the Liver. 2024;56(6):1064-1070. doi:10.1016/j.dld.2023.10.012.

68.

Prognostic Modeling of Overall Survival and Analysis of K-M Survival Curves in Patients With Primary Colon Cancer: A SEER-based Study.

Li C, Cao S, Sun X, Lu C, Guo M.

Medicine. 2023;102(23):e33902. doi:10.1097/MD.0000000000033902.

69.

Prognostic Nomograms for Predicting Cause-Specific Survival and Overall Survival of Stage I-Iii Colon Cancer Patients: A Large Population-Based Study.

Zhou Z, Mo S, Dai W, et al.

Cancer Cell International. 2019;19:355. doi:10.1186/s12935-019-1079-4.

70.

Predicting Survival After Curative Colectomy for Cancer: Individualizing Colon Cancer Staging.

Weiser MR, Gönen M, Chou JF, Kattan MW, Schrag D.

Journal of Clinical Oncology : Official Journal of the American Society of Clinical Oncology. 2011;29(36):4796-802. doi:10.1200/JCO.2011.36.5080.

71.

Accurate Prediction of Prognosis by Integrating Clinical and Molecular Characteristics in Colon Cancer.

Wang L, Su M, Zhang M, et al.

Frontiers in Cell and Developmental Biology. 2021;9:664415. doi:10.3389/fcell.2021.664415.

72.

Prognostic Factors and Survival Disparities in Right-Sided Versus Left-Sided Colon Cancer.

Asghari-Jafarabadi M, Wilkins S, Plazzer JP, Yap R, McMurrick PJ.

Scientific Reports. 2024;14(1):12306. doi:10.1038/s41598-024-63143-3.

73.

Survival Rates and Prognostic Factors in Right- And Left-Sided Colon Cancer Stage I-Iv: An Unselected Retrospective Single-Center Trial.

Degro CE, Strozynski R, Loch FN, et al.

International Journal of Colorectal Disease. 2021;36(12):2683-2696. doi:10.1007/s00384-021-04005-6.

74.

Prognostic Survival Associated With Left-Sided vs Right-Sided Colon Cancer: A Systematic Review and Meta-analysis.

Petrelli F, Tomasello G, Borgonovo K, et al.

JAMA Oncology. 2017;3(2):211-219. doi:10.1001/jamaoncol.2016.4227.

75.

Age-Related Differences in Colon and Rectal Cancer Survival by Stage, Histology, and Tumour Site: An Analysis of United States SEER-18 Data.

Pilleron S, Withrow DR, Nicholson BD, Morris EJA.

Cancer Epidemiology. 2023;84:102363. doi:10.1016/j.canep.2023.102363.

76.

Black-White Disparities Across the Colorectal Cancer Care Continuum in the USA.

May FP, Mehtsun WT, Jemal A, Gupta S.

Nature Reviews. Gastroenterology & Hepatology. 2025;:10.1038/s41575-025-01087-3. doi:10.1038/s41575-025-01087-3.

77.

Framework and Strategies to Eliminate Disparities in Colorectal Cancer Screening Outcomes.

Doubeni CA, Selby K, Gupta S.

Annual Review of Medicine. 2021;72:383-398. doi:10.1146/annurev-med-051619-035840.

78.

Navigating disparities: Insights from a colorectal cancer screening program in federally qualified community healthcare centers.

Amin R, Turner C, Santellano B, et al.

Journal of Clinical Oncology. 2024;42(Suppl 16):1537.
doi:10.1200/JCO.2024.42.16_suppl.1537.

Summary document 2:

symptoms and typical side effects from treatment in colon cancer patients across all ages and stages, including hereditary syndromes and high-risk populations, are the clinical practice guidelines from the American Society of Colon and Rectal Surgeons, the Delphi Initiative for Early-Onset Colorectal Cancer International Management Guidelines (DIRECt), and recent reviews in JAMA and NEJM.

[1-4]

The American Society of Colon and Rectal Surgeons guidelines detail that colon cancer may be asymptomatic or present with fatigue, blood in the stool, abdominal pain, changes in bowel habits, or obstructive symptoms. Advanced disease is more likely to be symptomatic, and hereditary syndromes (e.g., Lynch syndrome, FAP) may present earlier and with more aggressive features.

[1]

The Delphi Initiative (DIRECt), supported by the Italian Association for Inherited and Familial Gastrointestinal Tumors and other societies, recommends that symptom management—including pain, fatigue, nausea, vomiting, constipation, diarrhea, and cachexia—should follow ASCO and ESMO guidelines for all ages and risk groups. Enhanced antiemetic prophylaxis is advised for early-onset patients, especially women with low BMI, and early physical activity and nutritional support are beneficial for muscle mass maintenance.

[2]

Sexual dysfunction, infertility, and psychosocial issues are common and require proactive counseling, especially in younger and hereditary cases.

[2]

Treatment side effects vary by modality: surgery may cause bowel dysfunction and adhesions; chemotherapy (e.g., oxaliplatin, 5-FU) is associated with neuropathy,

myelosuppression, mucositis, and gastrointestinal symptoms; targeted therapies and immunotherapy can cause dermatologic, hepatic, and immune-related adverse events.

[3]

Table 2 from Lieberman et al. in JAMA provides a concise summary of follow-up and surveillance recommendations for higher-risk individuals, which is essential for monitoring and managing late effects.

Table 2. Colorectal Cancer Screening and Follow-up in Higher-Than-Average-Risk Individuals

Risk Level for CRC	Lifetime Risk of CRC	Initiation of Screening	Follow-up Intervals
Very high-germline mutations			
Familial polyposis	Nearly 100%	During teenaged years	Not applicable; colon resection is recommended
Lynch syndrome	Variable depending on mutation	Age 20-25 years	Colonoscopy every 2 years
High risk			
Ulcerative colitis or Crohn colitis	Increases with time 10%-20%	After 8 years of chronic disease	Colonoscopy every 2 years
Familial risk (first-degree relative ≤60 years)	10%-20%	Age 40 years or 10 years younger than index family member	Colonoscopy every 5 years
Familial risk (first-degree relative >60 years)	10%-15%	Age 40 years or 10 years younger than index family member	Any recommended screening test
High risk (personal history of colorectal neoplasia)			
Personal history of CRC	Uncertain; >10%	1 year after baseline colonoscopy with diagnosis of CRC	Colonoscopy at 3 years then again at 5 years if negative for neoplasia ; colonoscopy at appropriate interval if polyps are detected
High-risk adenoma(s) (adenoma ≥10mm with villous histology, high-grade dysplasia, or >3 adenomas or serrated polyps with dysplasia)	Uncertain; greater-than-average risk		Colonoscopy at 3 years after high-quality baseline colonoscopy with complete removal of polyps
Low-risk adenoma(s) (1-2 tubular adenomas <10 mm)	Uncertain; perhaps similar to average risk		Colonoscopy at 5-10 years after high-quality baseline colonoscopy with complete removal of polyps

Abbreviation: CRC, colorectal cancer.

Table 2. Colorectal Cancer Screening and Follow-up in Higher-Than-Average–Risk Individuals

Screening for Colorectal Cancer and Evolving Issues for Physicians and Patients: A Review. JAMA. November 21, 2016.

This table contextualizes the recommended intervals for colonoscopy and surveillance in high-risk populations, which is critical for early detection of symptoms and side effects in hereditary and high-risk patients.

References:

The American Society of Colon and Rectal Surgeons Clinical Practice Guidelines for the Management of Colon Cancer.

Vogel JD, Felder SI, Bhama AR, et al.

Diseases of the Colon and Rectum. 2022;65(2):148-177. doi:10.1097/DCR.0000000000002323.

Practice Guideline

2.

Delphi Initiative for Early-Onset Colorectal Cancer (DIRECt) International Management Guidelines.

Cavestro GM, Mannucci A, Balaguer F, et al.

Clinical Gastroenterology and Hepatology : The Official Clinical Practice Journal of the American Gastroenterological Association. 2023;21(3):581-603.e33. doi:10.1016/j.cgh.2022.12.006.

Practice Guideline

3.

Early-Onset Gastrointestinal Cancers.

Jayakrishnan T, Ng K.

JAMA. 2025;:2836671. doi:10.1001/jama.2025.10218.

Leading Journal New Research

4.

Increasing Incidence of Early-Onset Colorectal Cancer.

Sinicrope FA.

The New England Journal of Medicine. 2022;386(16):1547-1558. doi:10.1056/NEJMra2200869.

Leading Journal

5.

Screening for Colorectal Cancer and Evolving Issues for Physicians and Patients: A Review.

Lieberman D, Ladabaum U, Cruz-Correa M, et al.

JAMA. 2016;316(20):2135-2145. doi:10.1001/jama.2016.17418.

Leading Journal